Econ 5253 - Spring 2025
Problem Set 11 Rough Draft
Liverpool FC Performance Analysis

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1 Introduction

This project analyzes the performance of Liverpool FC in the 2023/24 season compared to the 2022/23 season. The goal is to evaluate whether the team has improved or regressed using performance metrics such as:

- Wins, draws, and losses
- Goals scored and goals conceded
- Expected goals (xG) and expected assists (xA)
- Possession statistics
- Shots, shots on target, and other team-level metrics

The analysis is relevant given the club's title ambitions and potential managerial transition at the end of the season.

2 Background

Football analytics increasingly relies on advanced metrics like xG and xA to assess team and player performance. Expected goals (xG) quantify the quality of scoring opportunities, while expected assists (xA) reflect pass quality leading to shots. These metrics help evaluate team efficiency and strategy beyond surface-level outcomes like wins or total goals.

Data will be gathered from well-known football analytics sites such as Understat and FBRef(**??**). Previous literature on expected goals will also inform the analysis(**?**).

3 Data

Sources

- **Understat**: match-by-match xG and team stats
- **FBRef**: seasonal summary statistics including possession and passing metrics

Time Frame

- 2022/23 season (baseline)
- 2023/24 season (current)

Metrics Collected

- Match outcomes (W-D-L)
- Goals scored/conceded
- xG, xA
- Shots per match, possession

4 Methods

- Clean and combine datasets from Understat and FBRef
- Calculate season averages per match for each metric
- Use visualizations (e.g., bar plots, line charts) to compare seasons
- Optional: regress team points or goal difference on xG metrics

5 Preliminary Results

The following table will compare Liverpool's key performance metrics across the two seasons:

Metric	**2022/23**	**2023/24 (YTD)**
Points per match	TBD	TBD
Goals per match	TBD	TBD
xG per match	TBD	TBD
xA per match	TBD	TBD
Possession %	TBD	TBD

Additional graphs and tables will be included in the final report once data cleaning and analysis are complete.

6 Next Steps

- Finish data scraping and preprocessing
- Generate visualizations for all key metrics
- Write final analysis and interpretation of results
- Explore more advanced modeling if time permits